

Backend Modules

A) Auth Module (/auth/)

- Signup, login, token refresh
- Password hashing, account recovery
- Role handling: LEARNER, TEACHER (Coordinator)

Key endpoints

- POST /auth/signup
- POST /auth/login
- POST /auth/refresh

Security requirements

- HTTPS/TLS, JWT authentication, secure password hashing.

B) Profile Module (/profile/)

1. Store onboarding inputs:
 - native language, other languages, region
 - proficiency level (A1–C2 style)
 - learning goals, daily time commitment, topics of interest
 - learning style
2. Provide user settings (accessibility prefs, language prefs)

Key endpoints

- GET /profile/me
- PUT /profile/me (update)
- POST /profile/onboarding (complete setup)
- GET /profile/preferences
- PUT /profile/preferences

C) Exercise Module (/exercise/)

1. Serve exercise content
2. Accept attempts, grade responses, update XP/level/streak
3. Adaptive difficulty:
 - maintain learner skill estimates by topic
 - select next questions based on proficiency + recent errors + time/accuracy signals.

Key endpoints

- GET /exercise/recommended (returns "today's recommended lesson")
- GET /exercise/lesson/{lessonId}
- GET /exercise/questions?topic=...&difficulty=...
- POST /exercise/attempts (submit answer(s))
- GET /exercise/progress (summary for learner dashboard)

Typical attempt flow

1. Client fetches questions
2. Client submits answers (POST /exercise/attempts)
3. Backend grades + updates:
 - accuracy, time spent
 - XP, streak, level
 - skill metrics per topic
4. Backend returns immediate feedback + next suggestion

D) AI Assistant Module (/assistant/)

- Translate
- Explain grammar / sentence breakdown
- Practice generation (extra examples/exercises based on weak areas)

Key endpoints

- POST /assistant/translate
- POST /assistant/explain
- POST /assistant/practice

Implementation notes

1. Treat AI calls as external dependency:
 - add request timeouts
 - rate limit per user
 - store minimal logs (prompt metadata, not sensitive content unless required)
2. Cache common translations/explanations where safe.

E) Analytics Module (/analytics/) – Teacher Dashboard

Teacher dashboard requirements include:

- learner progress tracking
- engagement prediction graphs
- exportable reports

Key endpoints

- GET /analytics/teacher/overview
 - total students, average progress, daily learning time
- GET /analytics/teacher/students
 - list students + summary stats + "at-risk" flags
- GET /analytics/teacher/students/{studentId}
 - timeline of performance, topic weaknesses, streak history
- GET /analytics/teacher/engagement
 - aggregated engagement metrics (and predictions if implemented)
- POST /analytics/teacher/reports/export
 - async export job → returns a download link when ready

Role-based access control

- Learners must not access teacher endpoints.
- Teachers can only access students they're assigned.

F) Notifications

Module Key endpoints

- POST /notifications/register-device (push token)
- GET /notifications/preferences
- PUT /notifications/preferences
- POST /notifications/send (teacher → student message, or internal use)

